

	Predicted	
	0	2
actual	7	21
		13

accuracy = $\frac{\text{total di}}{\text{total}}$

similar 10x10 any kind

When Accuracy is Misleading?

In case of Imbalanced dataset

	Yes	No
900		100

Ex Airport Passenger terrorist

	Predicted	
	1	0
Accuracy	0	99.99%
	0	99.99%

mean 99.99%

Wrong accuracy.

Lecture 6 Eigen Value & Eigen Vectors.

Let $A = [a_{ij}]_{m \times n}$ be a square matrix.

of order n

Consider the homogeneous system of equation

$$AX = \lambda X \text{ or } (A - \lambda I)X = 0$$

Where I is an identity scalar matrix of order n and λ is scalar.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

$$[A - \lambda I] = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)^2 - 6$$

$$|A - \lambda I| = \lambda^2 - 2\lambda - 5 = 0$$

Characteristic polynomial to characteristic equation

~~to~~ which gives zero soln?

$$\text{for glimpse } [A - \lambda I] \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

↑
eigen vector

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 3 & 2 - \lambda \end{bmatrix}$$

$$= \lambda^2 - 6\lambda + 5$$

$$= \cancel{\lambda^2 - 2\lambda - 3\lambda + 5} \quad \lambda^2 - 5\lambda - \lambda + 5$$

$$\cancel{\lambda(\lambda - 2) - 3(\lambda - 1)} \quad \lambda(\lambda - 5) - 1(\lambda - 5)$$

$$\lambda = 1, 5$$

$$(A - I)(x) = \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$3x_1 + 1x_2 = 0 \quad \text{"infinite soln"}$$

$$3x_1 + 1x_2 = 0$$

By Row reduction.
You can calculate

$$x_1 = k \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\text{or } \begin{aligned} \cancel{4x_1 + 1x_2} &= 0 & 3x_1 + 1x_2 &= 0 \\ \cancel{3x_1 + 2x_2} &= 0 & y &= -3x \end{aligned}$$

Similarly.

$$\begin{bmatrix} 4 - 5 & 1 \\ 3 & 2 - 5 \end{bmatrix} = 0$$

$$y = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$x - 3y = 0$$

$$x = y \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

2. Non zero solnⁿ $\Rightarrow$ Non-trivial solnⁿ $\Rightarrow$ Infinite solnⁿ

$\Rightarrow$ L.D vectors i.e all are equivalent statement

4. $(n-r)$ give the value of the number of L.I. solnⁿ / L.I. eigen vectors of above system

5. If there are 3 eigenvalue distinct & unique then the eigen vectors will be linearly independent

6. If by chance same then may or may not independent.

7. Roots of characteristic equation are known as eigen values or characteristic values, or eigen roots or characteristic roots or latent roots

8. Set consisting eigen vector corresponding to different eigen values L.I.

9. The set of eigen vector of the matrix A is called spectrum & largest value in it called spectral radius.

Algebraic & Geometric Multiplicity

Algebraic Multiplicity: No. of times a particular eigen value repeats is known as algebraic multiplicity

Eg. $\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$

Count basically $(\lambda - 1)^2 (\lambda - 3) = 0$

Geometric Multiplicity: No. of distinct eigen vector for a particular eigen value is called Geometric multiplicity

$AM \geq GM$

Properties of Eigen value & Eigen Vectors

- i. Eigen values of a hermitian matrix are all real
- ii. Eigen values of skew hermitian matrix are either 0 or purely imaginary
- iii. Eigen value of orthogonal matrix unit modulus.
 $AA^T = I$
- iv. for idempotent matrix $A^2 = A$ 0 or 1
- v. Unitary Matrix $AA^H = I$ unit modulus
- vi. Eigen values of an involutory $A^2 = I$ ± 1 or $\pm i$

7. If A & B two matrices same order A and B

$B = P^{-1} A P$. or similar have
Same Eigen values.

8. If λ^0 is an eigen value of A then

λ^{-1} is an eigen value of A^{-1}

$\lambda_1, \lambda_2, \dots, \lambda_n$ then Eigen value A^{-1}
are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$

9. Eigen value of $\text{adj } A = \left(\frac{|A|}{\lambda} \right)$ Product of values

10. The matrix A and its transpose A' has the
Same characteristic roots

11. If $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are the eigen values of
square matrix A then,

(i) Eigen values of KA are $k\lambda_1, k\lambda_2, \dots, k\lambda_n$

(ii) Eigen values of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$

(iii) Eigen values of A^2 are $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$

$-I = A^3$ are $\lambda_1^3, \lambda_2^3, \dots, \lambda_n^3$

12. Any two characteristic vector corresponding to two distinct characteristic roots of a unitary matrix / or real symmetric matrix A are orthogonal

Theorem

1. Product of eigen values of matrix is equal to determinant of that matrix

2. Sum of eigen values of a matrix is equal to trace of that matrix

$$(i) \lambda_1 \times \lambda_2 \times \dots \times \lambda_n = |A|$$

$$(ii) \lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = \text{trace of } (A)$$

Ques. $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\lambda = 1, 6$$

$$\lambda_1 + \lambda_2 = \underline{7}$$

$$\lambda_1 \lambda_2 = \underline{6}$$